ModelIt!

Student Activity Pack

How Do Rocks Tell Stories?

Reading Earth's History from Rocks and Layers

Grade 2 | NGSS 2-ESS1-1 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Have you ever picked up a rock and wondered how it got there? What did you notice about it?

2 Do you think all rocks are the same or are there different kinds? How are they different?

3 Draw what you think is inside a mountain — what would you see if you cut it in half?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Earth Event | Rock Type | Rock Layer Pattern

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Slow River

Earth Event set to slow-flowing river over many years

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Fast Volcano

Earth Event set to volcanic eruption

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Slow Wind Erosion

Earth Event set to wind blowing sand for thousands of years

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Fossils in Layers — Remains of ancient plants and animals found inside rock layers — they tell us what was alive at the time that layer formed

Weathering Speed — How fast the rock breaks down from weather — soft rock erodes quickly, hard rock lasts a long time

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How can a rock tell us a story about something that happened on Earth a long, long time ago?
Use evidence from your simulation to support your answers.

1 How is reading rock layers similar to reading pages in a book?

2 What is one thing a rock at your school could tell you about the past?

8 STEM Challenge: Build a Rock Layer Timeline

ModelIt!

YOUR ENGINEERING MISSION

Create a model showing rock layers that tell the story of three different Earth events over time.

THE SCENARIO

A museum needs a display that shows how rock layers record Earth's history. Your team will build a model with at least three layers, each representing a different Earth event, to teach visitors how scientists read rocks like history books.

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU USE FOR EACH LAYER — SAND, CLAY, GRAVEL, OR PEBBLES?
2. HOW WILL YOU MAKE EACH LAYER LOOK DIFFERENT TO REPRESENT DIFFERENT EARTH EVENTS?
3. HOW WILL YOU LABEL YOUR LAYERS SO VISITORS KNOW WHAT EACH ONE REPRESENTS?
4. HOW DOES YOUR MODEL SHOW THAT SOME EVENTS ARE FAST AND SOME ARE SLOW?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Geologists study rocks, minerals, and the Earth's history. They help find resources like water and minerals, predict earthquakes and volcanoes, and teach us about Earth's past. They earn $55,000–$100,000/year.